

IoT Based Pothole Detection & Alert System

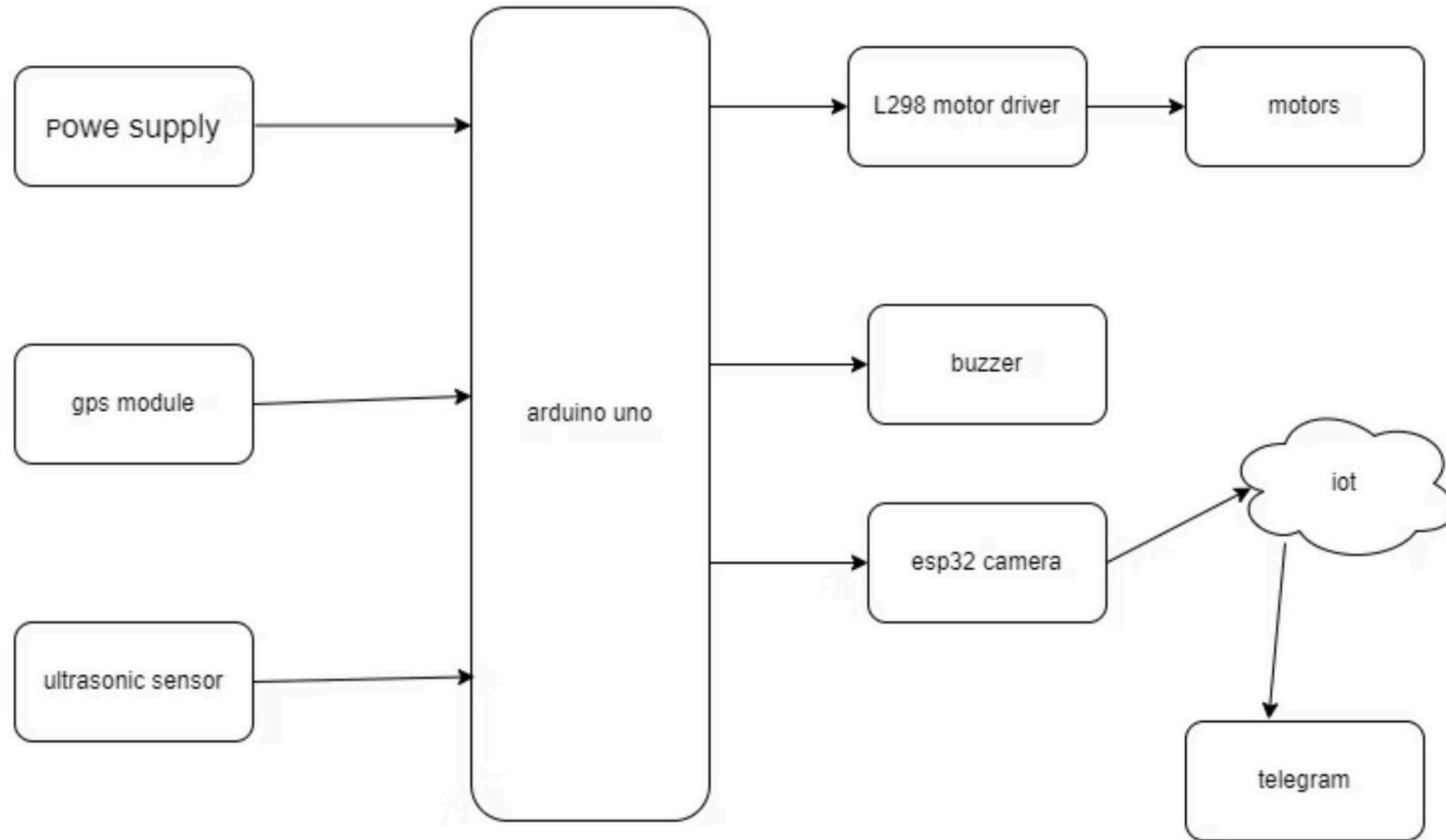
Abstract:

This project proposes an IoT-based Pothole Detection System designed for real-time monitoring and alerting of road potholes. Employing two ultrasonic sensors and GPS technology, the system ensures accurate detection and precise location tracking. Alerts are relayed to drivers through a mobile app using a buzzer and haptic vibrator. Additionally, the integration of the ESP32-CAM module allows for the capture of pothole images, which are transmitted to a Telegram channel along with GPS and ultrasonic data. The algorithm for pothole detection operates in real-time, and recorded data can be accessed by authorized individuals and drivers through a dedicated app or webpage. This system offers a comprehensive approach to road safety and maintenance, enhancing driver awareness and facilitating timely interventions.

Introduction:

- Road infrastructure is crucial for transportation, but potholes pose a significant threat to both vehicles and pedestrians. Traditional methods of pothole detection and reporting are often manual and time-consuming. The proposed system utilizes IoT technology to automate this process, ensuring swift and accurate detection of potholes.

Proposed system:



Hardware components:

- Arduino uno
- Ultrasonic sensor
- Gps module
- Motor driver
- Motors
- Esp8266 wifi module
- Jumper wires etc

Arduino Uno

.Arduino Uno is the main microcontroller used in the system.

- It controls all sensors and modules.
- It processes the data from ultrasonic sensor.
- It sends data to WiFi / GPS module.



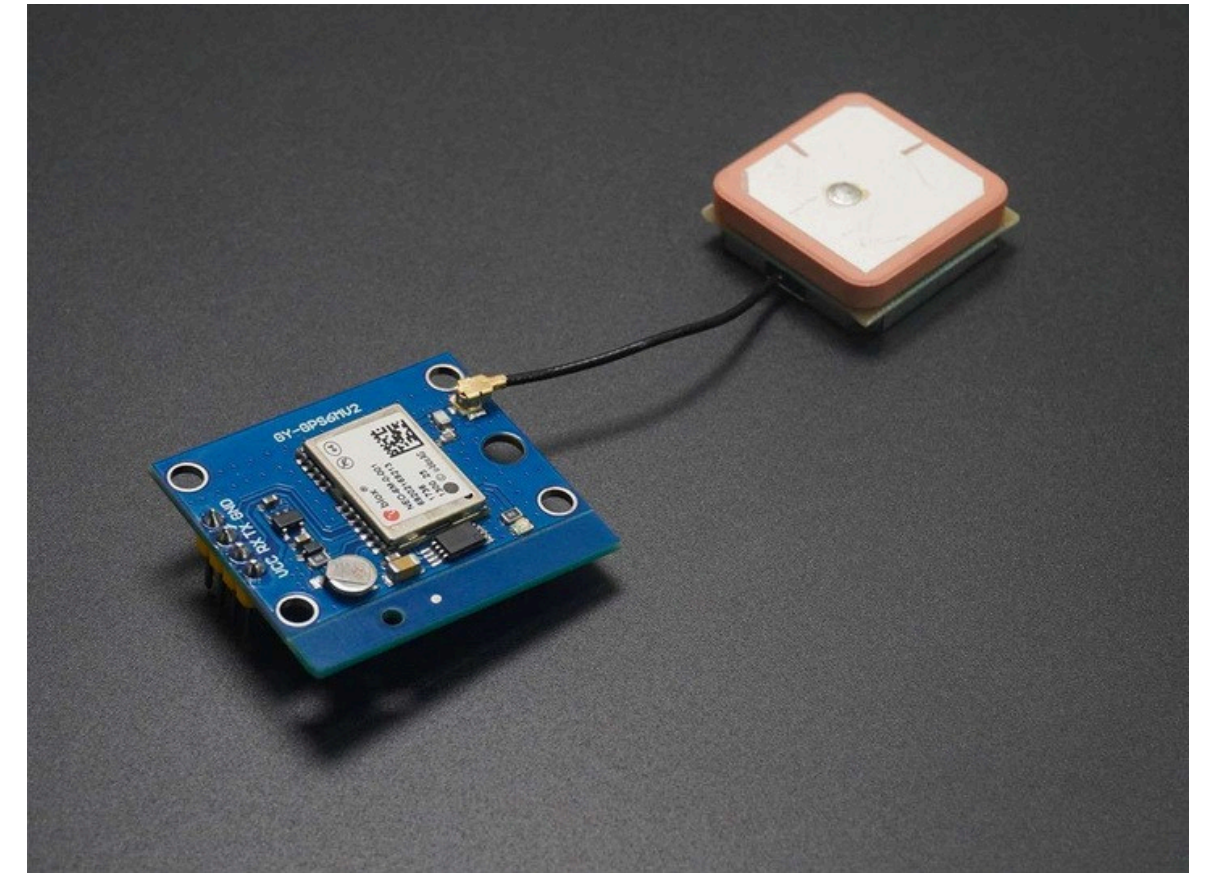
Ultrasonic Sensor

- Ultrasonic sensor is used to measure distance.
- It helps to detect potholes on the road.
- It sends distance data to Arduino.
- Used for obstacle / depth detection.



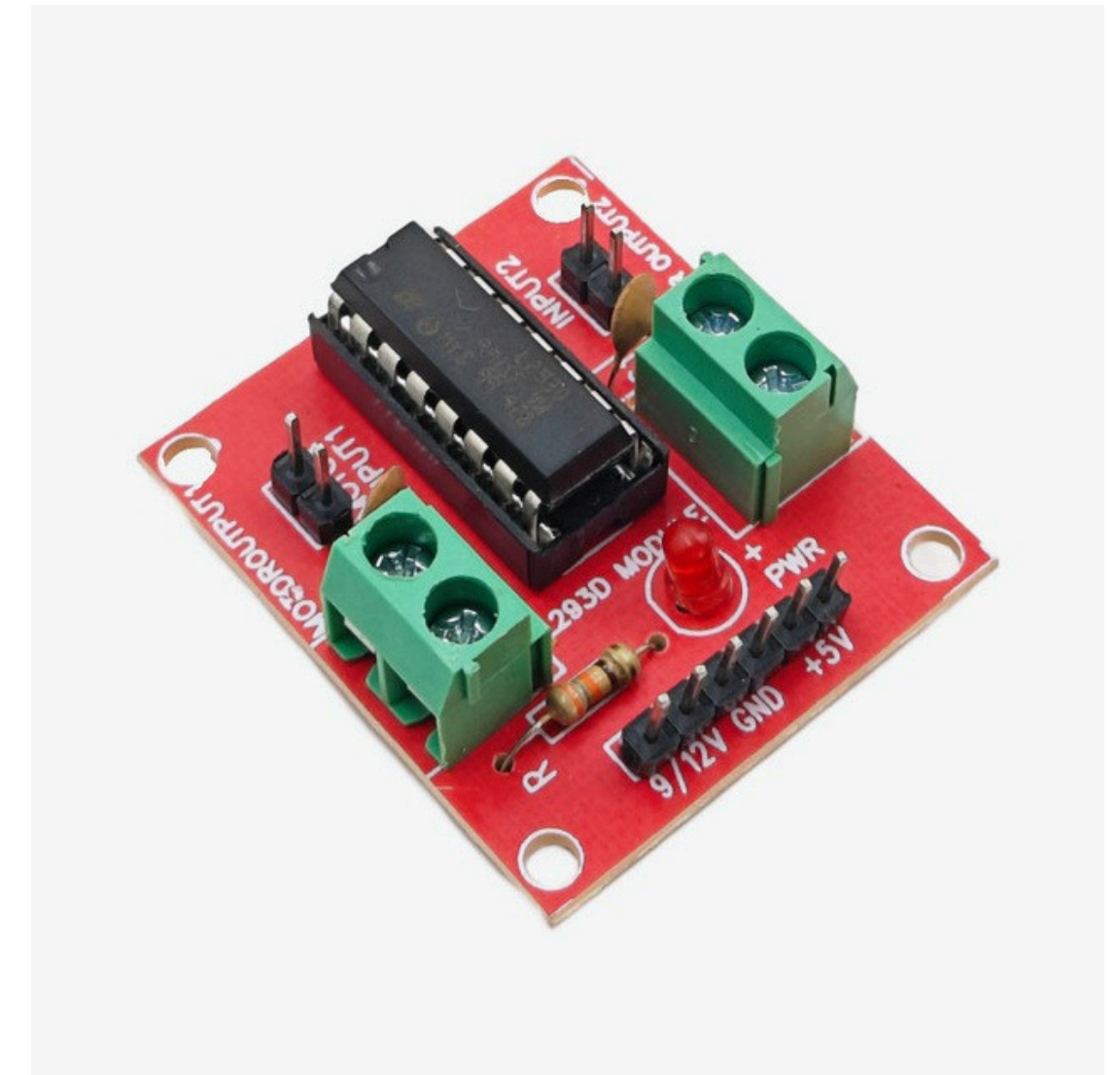
GPS Module

- GPS module is used to get location.
- It gives latitude and longitude.
- Used to send pothole location.
- Helps in tracking exact position.



Motor Driver

- It allows Arduino to run motors.
- Used to control speed and direction.
- Required for robot movement.
- Motor driver controls the motors.



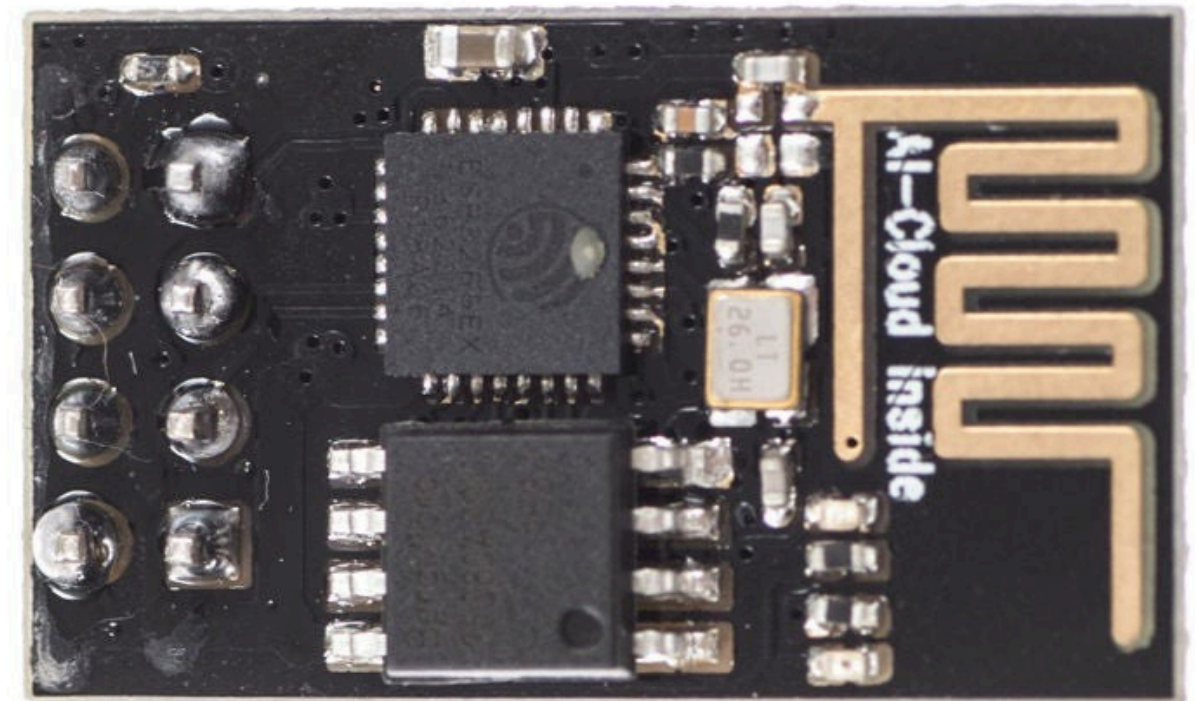
Motors

- Helps the robot move on road.
- Connected through motor driver.
- Motors are used for movement.
- Required for testing pothole detection.



ESP8266 Wifi Module

- ESP8266 is used for WiFi connection.
- Sends data to internet / server.
- Used for IoT communication.
- Helps in alert system.



Software components

- Arduino Software
- Arduino c language

Software Implementation

Requirement Analysis:

The system is designed to detect potholes using an ultrasonic sensor and obtain location using a GPS module. Arduino Uno controls all components, and ESP8266 WiFi module is used to send data through IoT. The program is written using Arduino IDE.

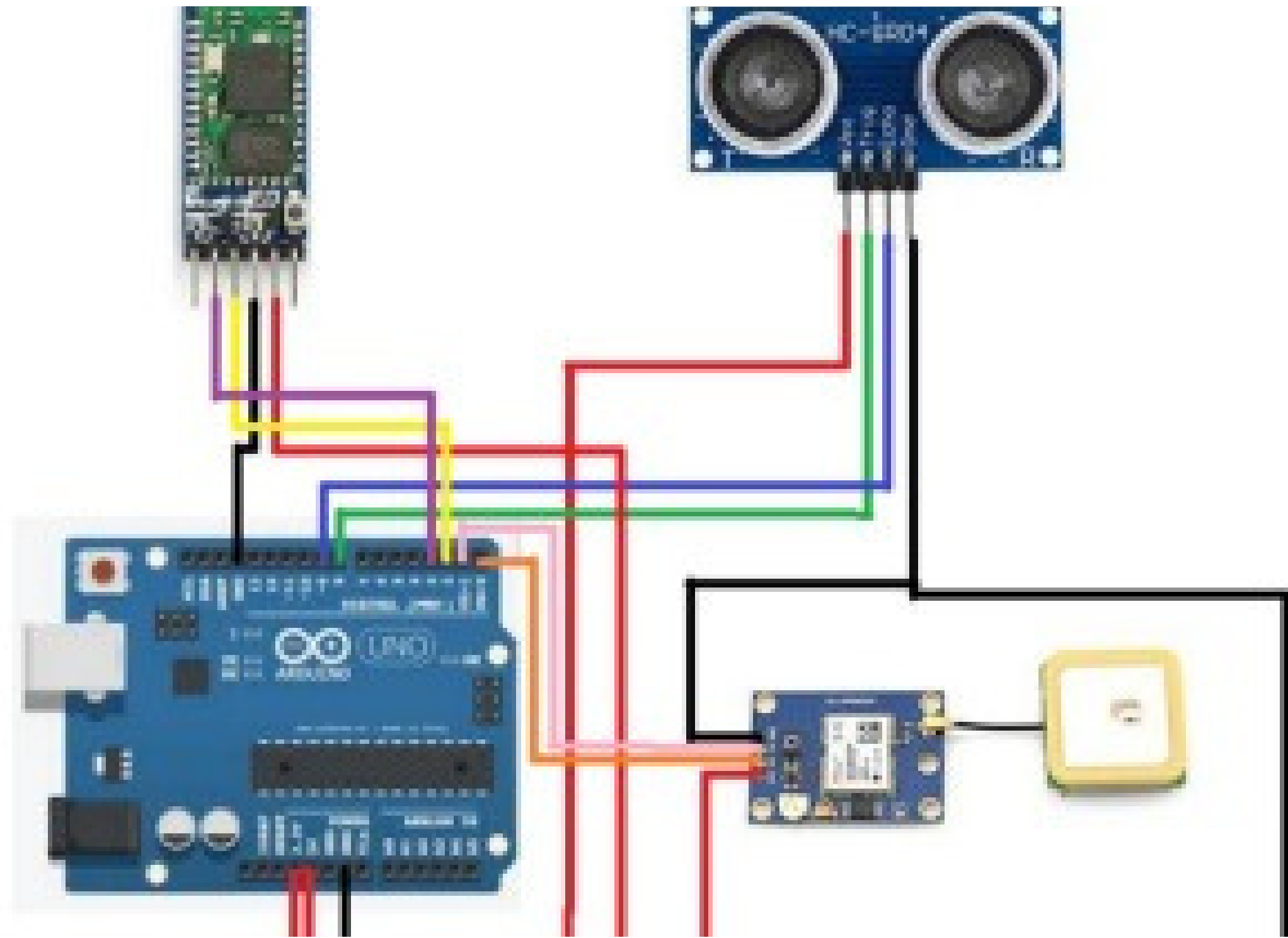
System Architecture:

The ultrasonic sensor sends distance data to Arduino. When a pothole is detected, the GPS module provides location coordinates, and ESP8266 sends the alert through WiFi. The motor driver controls the motors for movement.

Software Used

- Arduino IDE – used to write and upload code
- ESP8266 Library – for WiFi communication
- GPS Library – for location reading
- Embedded C / Arduino C – programming language
- Serial Monitor – for testing output

Circuit Diagram



Future Scope:

- 1. **Integration with Autonomous Vehicles:** Enhancing safety features in self-driving cars.
- 1. **Data Analytics for Predictive Maintenance:** Utilizing collected data to predict potential road issues.
- 1. **Global Implementation:** Expanding the system to address pothole concerns on a larger scale

Conclusion:

- TheloT-based PotholeDetecting and Alert System offers a technologically advanced and automated solution to the persistent issue of potholes in road infrastructure. By addressing detection delays and streamlining the repair process, the system contributes to overall road safety and maintenance.